MIST- MEDICAL INSTITUTE FOR SCREENING TEST

ANESTHESIA SUBJECT WISE TEST DATED ON 3rd MAY 2026

QUESTIONS

Q1. A 65-year-old man with a history of coronary artery disease underwent drug-eluting stent placement 1 year ago. He is currently taking aspirin and clopidogrel. He is now posted for elective cholecystectomy, and the surgeon is concerned about perioperative bleeding. What is the most appropriate management of clopidogrel before surgery?

a) Continue clopidogrel perioperatively
b) Stop clopidogrel 24 hours before surgery
c) Stop clopidogrel 5 days before surgery
d) Postpone surgery

Q2. According to modern preoperative fasting guidelines, how long should a patient fast from "clear fluids" (e.g., water, black coffee, apple juice) before elective surgery?

a) 6-8 hours
b) 4 hours
c) 2 hours
d) 12 hours

Q3. A 64-year-old man with diabetes mellitus and hypertension is planned for elective total hip arthroplasty. He sustained a myocardial infarction 1 year ago and now develops chest pain on climbing one flight of stairs and even at rest sometimes. According to the ASA Physical Status classification, he belongs to:

a) ASA III
b) ASA IV
c) ASA V
d) ASA VI

Q4. During a preoperative airway assessment, a patient is asked to protrude their tongue while seated, and only the hard palate is visible. How is this classified?

a) Class I
b) Class II
c) Class III
d) Class IV

Q5. A patient is scheduled for an elective surgery at 10:00 AM. According to standard NPO guidelines, when should they ideally stop consuming light meals (e.g., toast and tea)?

a) Midnight
b) 4:00 AM
c) 6:00 AM
d) 8:00 AM

Q6. Which of the following structures is visible in a Class II Mallampati score?

a) Soft palate, uvula, and pillars
b) Soft palate, fauces, and base of the uvula
c) Hard palate only
d) Soft palate and uvula, but pillars are obscured

Q7. A 65-year-old man in whom elective laparoscopic cholecystectomy under general anesthesia with controlled ventilation was being done.The patient remains hemodynamically stable, and ventilator parameters are unchanged. What is the most likely cause of this capnographic finding?

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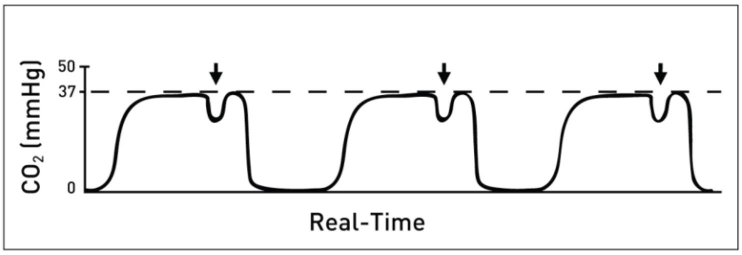
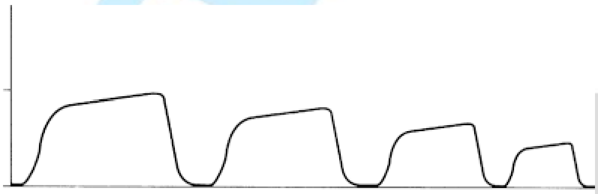


a) Pulmonary embolism
b) Inadequate neuromuscular blockade
c) Breathing circuit disconnection
d) Severe hypovolemia

Q8. A 34 yrs being operated for Laparoscopic cholecystectomy was being done, Pneumoperitoneum is created using carbon dioxide. Fifteen minutes after insufflation, a change in the end-tidal CO_2 waveform is noted as shown in the image below. Most probable cause :



a) CO2 embolism
b) Inadequate neuromuscular blockade
c) Breathing circuit disconnection
d) Severe hypovolemia

Q9. A 55-year-old man is undergoing an ultrasound-guided infraclavicular brachial plexus block using a highly protein-bound amide local anesthetic. Within minutes of injection, he develops anxiety, metallic taste, and circumoral numbness, followed by generalised muscle twitching. He remains conscious with BP 100/74 mmHg, pulse 100/min, and SpO_2 98% on supplemental oxygen. What is the most appropriate immediate management?
a) Administer 20% lipid emulsion therapy
b) Give intravenous magnesium sulfate
c) Start intravenous amiodarone
d) Administer calcium chloride

Q10. A 68-year-old man with long-standing diabetes mellitus has stage 4 chronic kidney disease and deranged liver function tests. He is scheduled for elective laparotomy under general anesthesia. Neuromuscular blocking agent of choice:
a) Vecuronium
b) Cisatracurium
c) Rocuronium
d) Pancuronium

Q11. The following image shows a portable inhalational analgesia cylinder used in the labour room. It provides rapid analgesia with quick recovery. What is the composition of the gas mixture shown?

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a) 70% N_2O + 30% O_2
b) 60% N_2O + 40% O_2
c) 50% N_2O + 50% O_2
d) 100% N_2O

Q12. A 45-year-old man undergoing surgery under general anesthesia is intubated, paralysed with a neuromuscular blocking agent, and maintained on intermittent positive pressure ventilation (IPPV). Which of the following Mapleson breathing circuits is most appropriate?
a) Mapleson A
b) Mapleson B
c) Mapleson C
d) Mapleson D

Q13. A patient undergoing surgery under general anesthesia is intubated after induction. Immediately after ventilation is started, confirmation of correct endotracheal tube placement is required. Which of the following is the most reliable method to confirm tracheal intubation?
a) Bilateral chest rise
b) Fogging inside the endotracheal tube
c) Visualisation of the tube in trachea on transesophageal echocardiography
d) Persistent end-tidal CO_2 waveform on capnography

Q14. A 58-year-old man with peritonitis is brought for emergency laparotomy. An induction agent is chosen that offers maximum cardiovascular stability but can transiently suppress adrenal steroid synthesis. Which drug was most likely used?
a) Propofol
b) Thiopentone
c) Etomidate
d) Midazolam

Q15. A 35-year-old man is taken for emergency laparotomy under general anesthesia. Rapid-sequence induction is performed with a high dose of a steroidal nondepolarizing neuromuscular blocker. Immediately after intubation, the procedure is cancelled due to an intraoperative concern, and the patient must be awakened. Neuromuscular monitoring shows no response to train-of-four stimulation. Which of the following is the most appropriate management for rapid reversal of neuromuscular blockade?
a) Neostigmine + glycopyrrolate
b) Sugammadex 2 mg/kg IV
c) Sugammadex 16 mg/kg IV
d) Flumazenil IV

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Q16. A 2-month-old infant undergoes repeated venipuncture in the NICU after application of a topical local anesthetic cream under an occlusive dressing. A few hours later, the infant becomes centrally cyanosed. What is the most likely cause?

a) Local anesthetic systemic toxicity due to lignocaine absorption
b) Methemoglobinemia
c) Hypoventilation
d) Allergy

Q17. A 7-year-old boy is brought to the emergency department with a displaced fracture of the radius and ulna requiring closed reduction. An intravenous anesthetic agent is administered for procedural sedation causing dissociative anesthesia :

a) Ketamine
b) Propofol
c) Etomidate
d) Thiopentone sodium

Q.18 Arrange the following inhalational anesthetic agents in order of increasing potentiation of neuromuscular blockade:

I. Halothane
II. Isoflurane
III. Sevoflurane
IV. Desflurane

a) iv > iii > ii > i
b) iii > iv > ii > i
c) ii > iii > iv > i
d) i > iii > iv > ii

Q .19. A 28-year-old woman is evaluated in the pre-anesthetic clinic before elective surgery. She is seated upright and asked to open her mouth fully and protrude her tongue without phonation. According to the Modified Mallampati classification, the airway shown corresponds to:

a) Grade I
b) Grade II
c) Grade III
d) Grade IV

Q.20. The below cylinder contains which gas?

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a) Oxygen
b) Nitrous oxide
c) Entonox
d) Heliox

Q21. A 45-year-old woman is undergoing elective abdominal surgery under general anesthesia. The anesthesiologist titrates anesthetic depth using a Bispectral Index (BIS) monitor. Which BIS value range best ensures an adequate level of general anesthesia for surgery?
a) 80–100
b) 60–80
c) 40–60
d) 20–30

Q22. A 30-year-old healthy woman is scheduled for elective laparoscopic surgery under general anesthesia at 10:00 AM. She drank a glass of clear apple juice at 6:30 AM but had no solid food after midnight. She is asymptomatic and vitals are stable. What is the most appropriate management?
a) Proceed with surgery as planned
b) Delay surgery for 4 hours
c) Delay surgery for 6 hours
d) Perform rapid sequence induction due to aspiration risk

Q23. A 70-kg male with ASA I status is posted for elective inguinal hernia repair under general anesthesia, for which he fasted overnight. After induction, he remains hemodynamically stable with adequate urine output and no blood loss. Which is the most appropriate intraoperative maintenance fluid?
a) No intravenous fluid needed
b) 5% dextrose infusion
c) Balanced crystalloid solution
d) 0.9% normal saline

Q24. A 58-year-old man suddenly collapses in the hospital corridor. A nurse immediately activates a Code Blue and reaches him within seconds. The patient is unresponsive. On assessment, he has no respirations with a palpable carotid pulse. What is the most appropriate immediate action by the first responder?
a) Start chest compressions
b) Provide rescue breaths at 1 breath every 6 seconds
c) Place the patient in lateral recovery position and monitor
d) Reassess breathing and pulse simultaneously for 10 seconds

Q25. Given below is an airway device used during general anesthesia. Consider the following statements:

I. Provides a higher oropharyngeal seal pressure
II. Contains a gastric drainage channel
III. Reduces the risk of gastric insufflation and aspiration
IV. Acts as a definitive airway with complete protection against aspiration
V. Requires direct laryngoscopy for insertion

Which of the following statements is/are correct?

a) i, ii, iii
b) ii, iv, v
c) i, ii, iii, iv
d) iii, iv, v

Q26. A 42-year-old patient undergoes a 7-hour intracranial tumor excision under total intravenous anesthesia. The anesthesiologist wants an opioid infusion that can be continued throughout surgery but allows immediate neurological assessment and extubation at the end of the procedure, with no accumulation even after prolonged infusion. Which of the following agents is most appropriate?

a) Fentanyl
b) Morphine
c) Remifentanil
d) Sufentanil

Q27. A 54-year-old man presents with abdominal distension and repeated bilious vomiting for 2 days. He is drowsy and unable to protect his airway. His vital signs are BP 108/68 mmHg, HR 118/min, RR 28/min, and SpO_2 90% on face mask oxygen. Abdominal radiograph shows multiple air-fluid levels suggestive of acute intestinal obstruction. Rapid sequence intubation is planned. The patient has been monitored, intravenous access secured, suction kept ready, and adequately pre-oxygenated with 100% oxygen. What is the next step in management?

a) Apply cricoid pressure and perform laryngoscopy
b) Give a sedative and wait before intubation
c) Administer an induction agent followed immediately by a neuromuscular blocker
d) Start positive pressure ventilation with bag-mask device

Q28. During general anesthesia, the anesthesiologist uses a specialized vaporizer that is heated and pressurized and injects anesthetic vapor directly into the fresh gas flow instead of using a conventional variable-bypass vaporizer. Which inhalational anesthetic agent is being administered?

a) Desflurane
b) Sevoflurane
c) Halothane
d) Isoflurane

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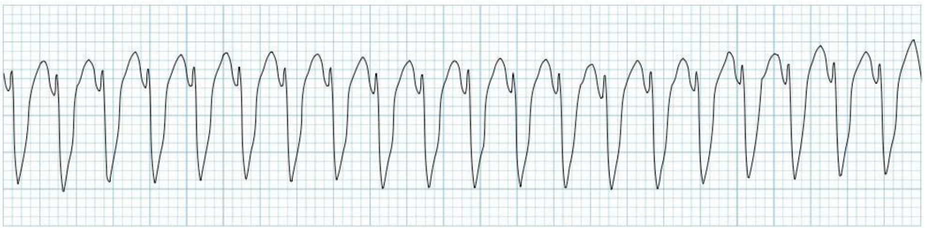
Q29. A 52-year-old woman is scheduled for cataract extraction under short general anesthesia as a day-care procedure. The procedure is expected to last 30–40 minutes and pre-anesthetic assessment is acceptable. She lives about 45 minutes from the hospital but states that she lives alone and will not have anyone to stay with her after discharge. Which factor makes her unsuitable for day-care surgery?

a) Living 45 minutes away from the hospital
b) Short duration of anesthesia
c) No responsible adult available at home after discharge
d) ASA physical status III

Q30. 19-year-old previously healthy male undergoes emergency appendicectomy under general anesthesia. Soon after induction and intubation, the anaesthetist notices increasing resistance to ventilation. Capnography shows a steadily rising end-tidal CO_2 despite increasing minute ventilation. The patient becomes tachycardic, develops generalized muscle rigidity, and his core temperature rises to 39.2°C. Arterial blood gas reveals metabolic acidosis. What is the most appropriate immediate management?

a) Administer intravenous dantrolene
b) Increase ventilation and administer 100% oxygen
c) Start active cooling with ice packs
d) Give intravenous sodium bicarbonate

Q31. A 62-year-old man with a history of coronary artery disease suddenly collapses in the emergency department triage area. He is unresponsive, not breathing normally, and no carotid pulse is felt. A cardiac monitor is immediately attached and shows the ECG tracing below. What is the most appropriate immediate action?



Arrange the following steps in the correct sequence of management:

I. Resume high-quality CPR for 2 minutes
II. Defibrillation
III. Reassess rhythm and pulse
IV. Administer intravenous epinephrine

a) ii → i → iii → iv
b) i → ii → iv → iii
c) ii → iii → i → iv
d) i → ii → iii → iv

Q32. An unconscious adult is brought to the emergency room after a syncopal episode. He is not responding to verbal commands and is breathing inadequately, but a carotid pulse is palpable. The healthcare provider performs the maneuver shown in the image. What is the primary purpose of this maneuver?

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a) To prevent aspiration of gastric contents
b) To relieve upper airway obstruction by the tongue
c) To stimulate respiratory effort
d) To improve lung compliance

Q33. A 64-year-old man with triple-vessel coronary artery disease is scheduled for elective CABG with cardiopulmonary bypass. He has no history of drug allergy. Which of the following is the most appropriate antibiotic prophylaxis regimen?
a) Erythromycin given 2 hours after skin incision
b) Vancomycin started after sternotomy closure
c) Cefazolin administered within 60 minutes before skin incision
d) Piperacillin–tazobactam for 5 days postoperatively

Q34. A 30-year-old woman is planned for lower limb surgery under lumbar epidural anesthesia at the L3–L4 interspace using a midline approach. As the Tuohy needle is advanced, the anesthesiologist identifies the loss of resistance and stops advancing further. Which of the following represents the correct order of structures pierced by the needle from superficial to deep until the epidural space is reached?
a) Skin → Subcutaneous tissue → Supraspinous ligament → Interspinous ligament → Ligamentum flavum → Epidural space
b) Skin → Subcutaneous tissue → Interspinous ligament → Supraspinous ligament → Ligamentum flavum → Epidural space
c) Skin → Subcutaneous tissue → Supraspinous ligament → Ligamentum flavum → Interspinous ligament → Epidural space
d) Skin → Subcutaneous tissue → Supraspinous ligament → Interspinous ligament → Dura mater

Q35. A 66-year-old woman with diabetes mellitus and hypertension is scheduled for elective total hip replacement. She becomes breathless while climbing a single flight of stairs and cannot walk more than 100 meters (functional capacity <4 METs). Resting ECG shows nonspecific ST-T changes. The anesthesiologist wants an investigation that best predicts the risk of perioperative major adverse cardiac events. Which of the following is the most appropriate investigation?
a) Exercise ECG testing
b) Dobutamine stress echocardiography
c) Myocardial perfusion scintigraphy
d) Coronary angiography

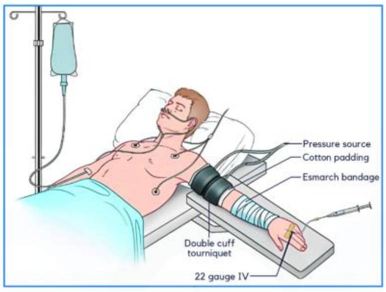
Q36. Regarding thiopentone sodium used for induction of anesthesia, consider the following statements:
I. Thiopentone produces rapid induction because of its high lipid solubility.
II. Its brief duration of action is due to rapid elimination from the body.
III. Thiopentone provides effective analgesia during anesthesia.
IV. The drug exerts its effect by enhancing activity at the GABA-A receptor.
V. It is mainly metabolized in the kidneys.

Which of the above statements is/are not correct?
a) i, iv
b) i, ii, iii
c) ii, iii, v
d) iii, iv, v

Q37. During general anesthesia, a 35-year-old patient receives atracurium for muscle relaxation. Within minutes, the anesthesiologist notices sudden hypotension, tachycardia, facial flushing, and bronchospasm. Oxygenation is adequate, and there is no rash suggestive of anaphylaxis. What is the most likely mechanism for these findings?
a) Cholinergic crisis due to acetylcholinesterase inhibition
b) Histamine release from mast cells
c) Malignant hyperthermia due to ryanodine receptor activation
d) Hyperkalemia due to depolarization at neuromuscular junction

Q38. A patient with acute COPD exacerbation is receiving oxygen via a Venturi mask set to deliver 24% oxygen. Suddenly the patient becomes tachypneic with a respiratory rate of 34/min and deep inspiratory efforts. The junior doctor worries that the patient will now receive a higher FiO_2 because of increased inspiratory flow. What will most likely happen to the delivered FiO_2?
a) It will increase significantly
b) It will decrease due to dilution
c) It will remain essentially unchanged
d) It will fluctuate with each breath

Q39. A patient is posted for carpal tunnel release surgery under the anesthesia technique shown below. Which of the following local anesthetic drugs is contraindicated in this technique?



a) Lignocaine
b) Prilocaine
c) Bupivacaine
d) Chloroprocaine

Q40. A 47-year-old man collapses in the hospital waiting area. Cardiac arrest is confirmed and CPR is started. Which of the following actions is most consistent with high-quality CPR in adults?
a) Compression rate 60–80/min with depth 3 cm
b) Frequent pauses to check pulse every 30 seconds
c) Compression rate 100–120/min with depth 5–6 cm and full chest recoil
d) Deliver 5 rescue breaths before starting chest compressions

Q41. This airway device is most appropriately indicated in which of the following clinical situations?

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a) Elective short surgical procedures under general anesthesia
b) Pediatric airway management
c) Thoracic surgery requiring one-lung ventilation
d) Postoperative oxygen supplementation in the recovery room

Q42. A 30-year-old woman is posted for lower segment cesarean section under spinal anesthesia. The anesthesiologist palpates the patient's back and draws a horizontal line connecting the highest points of both iliac crests to identify the appropriate intervertebral space for needle insertion. This anatomical landmark is called:
a) Reid's line
b) Tuffier's line
c) Spinous process line
d) Posterior axillary line

Q43. A 42-year-old obese woman undergoes elective surgery under general anaesthesia using an inhalational agent. About 10 days later, she develops fever, malaise, anorexia and progressive jaundice. Laboratory evaluation shows markedly elevated serum transaminases. Which of the following anaesthetic agents is the most likely cause?
a) Sevoflurane
b) Isoflurane
c) Halothane
d) Desflurane

Q44. A patient undergoing elective abdominal surgery under general anaesthesia receives a non-depolarising neuromuscular blocking agent. A peripheral nerve stimulator is used to assess the depth of blockade and recovery. Which of the following nerve–muscle units is most commonly used for neuromuscular monitoring?
a) Ulnar nerve – Adductor pollicis
b) Facial nerve – Orbicularis oculi
c) Posterior tibial nerve – Flexor hallucis brevis
d) Facial nerve - Corrugator supercilii

Q45. A 62-year-old man is undergoing cardiopulmonary resuscitation for cardiac arrest. He is intubated and connected to continuous waveform capnography. For several minutes, the end-tidal CO_2 ($EtCO_2$) values remain around 10 mmHg despite ongoing chest compressions. Suddenly, the $EtCO_2$ abruptly increases to 40 mmHg without any change in ventilation or compression technique. This finding most likely indicates:
a) Esophageal intubation
b) Hyperventilation by the rescuer

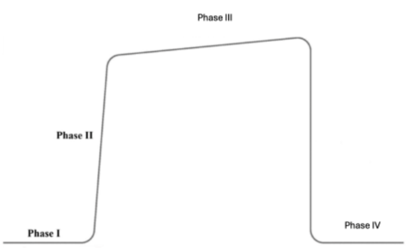
c) Return of spontaneous circulation (ROSC)
d) Pulmonary embolism

Q46. During controlled mechanical ventilation, an anesthesiologist replaces a 7.0 mm internal diameter endotracheal tube (ETT) with a 6.0 mm tube in the same patient. Which of the following is the most likely effect on airway resistance?
a) Resistance decreases slightly
b) Resistance remains unchanged
c) Resistance increases markedly
d) Resistance becomes half

Q47. A 64-year-old man with septic shock has been on invasive mechanical ventilation for 14 days. Because prolonged ventilatory support is anticipated, a surgical tracheostomy is performed. The patient continues to require positive-pressure ventilation with PEEP and has copious oropharyngeal secretions. Which tracheostomy tube is most appropriate?
a) Uncuffed tracheostomy tube
b) Cuffed tracheostomy tube
c) Jackson tracheostomy tube
d) Fenestrated tracheostomy tube

Q48. A 25-year-old man is brought to the emergency department after a high-speed road traffic accident. He is restless, pale, tachycardic (HR 138/min) and hypotensive (BP 70/40 mmHg). Active bleeding is noted from the thigh. Rapid transfusion of crystalloids and blood products is planned. Which intravenous access is most appropriate for initial resuscitation?
a) 26-gauge peripheral cannula
b) 16-gauge peripheral cannula
c) 14-gauge peripheral cannula
d) Central venous catheter

Q49. A capnogram obtained during general anaesthesia is shown above with phases labeled I–IV. Which of the following statements is not true?



a) Phase I represents dead space ventilation
b) End-tidal CO_2 is measured at the end of Phase III
c) CO_2 concentration is zero during Phase III
d) Phase IV corresponds to inspiration

Q50. The device shown in the image is being used to provide oxygen to a trauma patient in the emergency room. Which of the following statements regarding this device is correct?

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a) A low-flow device delivering $FiO_2 < 40\%$
b) Requires oxygen flow of at least 10–15 L/min to prevent reservoir bag collapse
c) Delivers a fixed and precise FiO_2 independent of patient's breathing pattern
d) It works by entrainment of room air through calibrated jets

Q51. During induction of anesthesia, immediately after injecting a drug through a cannula placed at the wrist, the patient complains of sudden severe burning pain in the hand, and the fingers become pale with delayed capillary refill. Which of the following actions should not be done?
a) Leave the cannula in situ
b) Inject 10 mL of 1% lignocaine through the cannula
c) Administer heparin and start systemic anticoagulation
d) Apply ice packs and elevate the limb

Q52. A 26-year-old man is brought to the emergency department after a high-speed road-traffic accident. On evaluation, he is unconscious and has stridor with gurgling respirations. Examination reveals extensive mid-face fractures with profuse oral bleeding and multiple avulsed teeth. Bag-mask ventilation is attempted, but chest rise is inadequate, and oxygen saturation remains 80%. What is the next best step in airway management?
a) Nasopharyngeal airway insertion
b) Endotracheal intubation
c) Emergency tracheostomy
d) Emergency cricothyrotomy

Q53. A 58-year-old man is resuscitated from out-of-hospital cardiac arrest due to ventricular fibrillation. Return of spontaneous circulation (ROSC) is achieved after 12 minutes of CPR. He remains unconscious (GCS 5/15) and is intubated and mechanically ventilated. Blood pressure is stable. What is the next most important step in post-resuscitation management to improve neurological outcome?
a) Immediate extubation
b) Therapeutic targeted temperature management (32–36 °C)
c) Routine prophylactic anticonvulsants
d) Hyperventilation to $PaCO_2 < 25$ mmHg

Q54. The Jackson–Rees breathing circuit is most appropriately used in which of the following situations?
a) Adult patient undergoing craniotomy with controlled ventilation
b) Two years old undergoing inguinal hernia repair under general anesthesia
c) Adult patient undergoing coronary artery bypass graft surgery
d) Parturient undergoing cesarean section under general anesthesia

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Q55. A 2-year-old child is brought to the emergency room after sudden coughing and breathing difficulty while playing with peanuts. On arrival, the child is cyanosed and becomes unresponsive. No effective breathing is seen. What should be done immediately?

a) Attempt Heimlich maneuver
b) Begin chest compressions and CPR
c) Perform blind finger sweep
d) Give back blows

ANSWERS

1	c	12	d	23	c	34	a	45	c
2	c	13	d	24	b	35	b	46	c
3	b	14	c	25	a	36	c	47	b
4	d	15	c	26	c	37	b	48	c
5	c	16	b	27	c	38	c	49	c
6	d	17	a	28	a	39	c	50	b
7	b	18	a	29	c	40	c	51	d
8	a	19	c	30	a	41	c	52	d
9	a	20	a	31	a	42	b	53	b
10	b	21	c	32	b	43	c	54	b
11	c	22	a	33	c	44	a	55	b